

Deflate by Bits, Not Trials: Why Query-Counted Deflated Sharpe Ratios Fail Under Adaptive Search, and a Sealed-Holdout Remedy

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Abstract

The Deflated Sharpe Ratio (DSR) is the standard defence against backtest overfitting. It raises the significance bar with the *number* of strategies tried, and recent work proposes logging every trial so that DSR can be applied to the true search intensity. We show that counting the backtests actually run is the wrong currency whenever the researcher is *adaptive*, that is, whenever each new strategy is chosen after seeing earlier validation results. This describes everyday quantitative practice, and it describes LLM research agents and RL alpha miners by construction.

First, we prove that a one-line “combine what worked” procedure over k pure-noise strategies drives the validation t -statistic to $\sqrt{2k/\pi} (1 + o_p(1))$, independently of cross-strategy correlation. The DSR bar for the same complete ledger grows only as $\sqrt{2 \ln k}$, so DSR certifies noise with probability tending to one. Second, we prove a matching positive result. If the holdout can reveal only a transcript from a set $\mathcal{T}$, then certifying at $t > \bar{G}^{-1}(\alpha/|\mathcal{T}|)$ is valid for *every* adaptive analyst. DSR is the non-adaptive special case $|\mathcal{T}| = N$, and overfitting scales as $\Theta(\sqrt{\ln(\text{bits revealed})})$ rather than $\sqrt{\ln(\text{queries})}$. Under full-precision Sharpe feedback, the only generic valid correction is the classical span (Scheffé/GRS) bound, which grows as $\sqrt{\dim}$ and is unusable at realistic scale. This motivates a *sealed holdout*: an opaque type whose only operation is a budgeted PASS/FAIL verdict.

On synthetic markets, and on 423 real industry strategies (1970–2026) under a stationary-bootstrap null, ordinary adaptive workflows such as “rank on validation, keep the top j ” are certified by query-counted DSR on 84–100% of pure-noise replicates, and sealed-verdict variants hold false certification at 0.8–3.0%. With frontier code-executing agents (Claude Code), the outcome depends on the instruction. Asked for a robust strategy, the agents usually decline to tune on the holdout. Instructed to maximise validation Sharpe, Claude Opus autonomously executes our attack in about a minute: it sweeps the single signals and then sign-weights the strongest ones, reaching validation t of up to 15.7 on pure noise, against a test t of about 0. Ledger DSR certifies these searches at the attack step under optional stopping (4 of 12 episodes, against 0 of 12 for a non-adaptive baseline) and is rescued only when continued search inflates its variance estimate. A GRPO alpha miner with an AlphaGen-style pool reward learns the same ensemble from reward alone in one of three seeds (177 signals, validation $t = 12.0$, test $t = -0.56$), and ledger DSR certifies it. In research campaigns that reuse one holdout across rounds, a researcher who builds on earlier full-precision results makes a false discovery in 100% of 5-round null campaigns, whether certifying by a Bonferroni bar or by cumulative-ledger DSR; splitting the holdout instead cuts power to 0.03 at 10 rounds. The sealed verdict stays at 1–3% while adapting, and keeps power of 0.65–0.79.

1 Introduction

A backtest is a hypothesis test whose researcher can see the answer key. Finance has long known that searching over many strategies and reporting the best manufactures significance [15, 16, 27, 28]. The modern standard remedy is to *count the search*. The Deflated Sharpe Ratio [1] raises the significance bar with the number of trials N , roughly as $\sqrt{2 \ln N}$ standard errors. The Probability of Backtest Overfitting [2] and reality-check bootstraps [13, 22, 28] likewise treat the search as a fixed family of candidates.

Real research is not a fixed family. A researcher who sees that momentum in banks and in insurers both validated well will try them together next; one who sees a signal fail will flip or drop it. Each step is innocuous, yet the candidate list is being written *by* the validation set. The question is acute now that this loop is being automated. LLM agents read a backtest and write the next hypothesis, and RL miners optimise a validation metric directly [26, 30]. Two recent responses illustrate the state of the art. Gençay [11] give an LLM discovery system a complete *trial ledger* so that DSR can be indexed to the true search intensity. Qu et al. [20] score factors only on post-submission (live) data with anytime-valid e-processes, paying roughly 500 trading days of waiting per verdict. The first assumes that counting trials suffices for historical backtests. The second gives up on historical backtests.

This paper. We show that for adaptive researchers the right currency is *information*, not trials. We make this precise for Sharpe-ratio certification and test it empirically.

1. Complete-ledger DSR is invalid under adaptive search (Theorem 1, Corollary 2).

An analyst who queries k pure-noise strategies, goes long the above-median ones and short the rest obtains a validation t -statistic of $\sqrt{2k/\pi}(1 + o_p(1))$ for any common correlation. DSR fed the complete $(k+1)$ -trial ledger certifies it with probability $\rightarrow 1$. Empirically, the everyday “keep the top- j by validation Sharpe” workflow is DSR-certified on 95% of synthetic and 84% of real-data null replicates, and greedy combination on 97–100%.

2. Information budgets give valid certificates (Theorem 3).

If everything the analyst can learn about the holdout is a transcript from a set $\mathcal{T}$, the bar $\bar{G}^{-1}(\alpha/|\mathcal{T}|)$ controls false certification for *any* analyst, however adaptive. DSR’s $\sqrt{2 \ln N}$ is the special case of $\ln N$ nats. The $\sqrt{\text{bits}}$ rate is attained up to constants (Proposition 6), and with full-precision feedback the only generic alternative is the classical span bound, which pays $\sqrt{\text{dim}}$ (Proposition 5).

3. A sealed holdout (Section 4).

We turn the theory into an interface: the holdout is an opaque type whose only eliminator is `verdict(strategy) -> PASS|FAIL`, with a query budget K and a fixed bar $\bar{G}^{-1}(\alpha/K)$. Validity holds by construction for any proposer, human, scripted, LLM or RL, without trusting or even logging it.

4. Evidence (Section 5).

We give theory-matching simulations (E1); a validity and power study including regime rotation (E2); 423 real hedged industry strategies with a stationary-bootstrap null (E3); exploratory studies of open-weight LLM agents (E4), which reproduce the attack’s behaviour but not its scale, and of GRPO alpha miners (E5), one of which learns the full ensemble attack and is DSR-certified on noise; frontier code-executing agents (E6), where Claude Opus, instructed to optimise, executes the attack autonomously and defeats ledger DSR under optional stopping, while the same agents usually decline to overfit when asked for robustness; and multi-round campaigns that share one holdout (E7), where the sealed verdict is the only protocol that is both reusable and valid for a researcher who learns from the holdout.

What is and is not new. Adaptive data analysis is a mature area of theoretical computer science and statistics [3, 5, 6, 14, 23, 29]. Our attack is a Sharpe-ratio relative of Freedman’s paradox [7] and of the boosting attack on ML leaderboards [4]. Theorem 3 is a high-probability transcript-counting bound in the spirit of description-length arguments [5] and mutual-information bounds [23]. Span bounds for the maximum Sharpe ratio are classical [12, 24]. Our contributions are (i) to show, formally and on real data, that the instrument now proposed for automated research, query-counted ledger DSR, fails under adaptive search; (ii) to identify DSR as the $\ln N$ -nat special case of an information-budget certificate, which yields a drop-in valid replacement; (iii) to package that replacement as a typed interface that automated researchers can be given safely; and (iv) to measure how far current LLM and RL researchers are from exploiting the gap. To our knowledge (i)–(iv) have not appeared in the finance or LLM-agent literature.

2 Setup

A holdout $H = (r_t)_{t=1}^n$ consists of n periods of returns on k base strategies, $r_t \in \mathbb{R}^k$. A strategy is a weight vector $w \in \mathbb{R}^k$ with return series $w^\top r_t$, per-period Sharpe ratio $\widehat{\text{SR}}_H(w) = \bar{x}/s_x$, and t -statistic $t_H(w) = \sqrt{n} \widehat{\text{SR}}_H(w)$.

Null. Under H_0 the rows r_t are i.i.d. with mean zero. For any w chosen independently of H , $t_H(w)$ is exactly Student- t_{n-1} for Gaussian rows and asymptotically standard normal in general. Write $\bar{G}(c) = \mathbb{P}(t_{n-1} > c)$.

Analysts and channels. An analyst (human, LLM agent or RL policy) interacts with H only through a *channel*. On the j -th query w_j the channel returns an answer a_j , and the transcript is $\pi = (a_1, a_2, \dots)$. Everything else the analyst uses is collected into U : training data, model weights, prompts, sampling randomness. We assume U is independent of H . The analyst reports $\hat{w} = f(U, \pi)$ and the holdout *certifies* it if $t_H(\hat{w})$ exceeds a bar. Validity means $\mathbb{P}_{H_0}(\text{certify}) \leq \alpha$. The status-quo channel returns $a_j = \widehat{\text{SR}}_H(w_j)$ in full precision.

DSR on a ledger. Given a ledger of N trials whose holdout Sharpe estimates have cross-sectional variance V , DSR [1] is

$$\text{DSR} = \Phi \left(\frac{(\widehat{\text{SR}} - \widehat{\text{SR}}_0) \sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{\text{SR}} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{\text{SR}}^2}} \right), \quad \widehat{\text{SR}}_0 = \sqrt{V} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right], \quad (1)$$

with γ the Euler–Mascheroni constant. A strategy is certified when $\text{DSR} > 0.95$. The bracket approximates $\mathbb{E} \max$ of N i.i.d. standard normals, $\approx \sqrt{2 \ln N}$. *Ledger conventions.* Throughout, N counts *distinct* strategies evaluated on the holdout, and V is the sample variance of their holdout Sharpe ratios; the reported strategy is added to the ledger if absent. In E5’s pool runs the ledger also contains every pool state, which only makes DSR more conservative. Under optional stopping (E6) the ledger is the set of distinct strategies seen up to the stopping time, with at least 10 entries.

3 Theory

3.1 Query counting fails under adaptivity

Theorem 1 (Adaptive ensemble attack). *Let the rows of H be i.i.d. $\mathcal{N}(0, \Sigma)$ with $\Sigma = (1 - \rho)I_k + \rho\mathbf{1}\mathbf{1}^\top$, $\rho \in [0, 1)$, and k even. The analyst queries each base strategy once, observes $Z_i = t_H(e_i)$, and reports*

$$\hat{w} = \frac{1}{k}s, \quad s_i = \text{sign}(Z_i - \text{med}(Z)) \quad (\text{rank long-short}).$$

Then for fixed k , as $n \rightarrow \infty$, $t_H(\hat{w}) \Rightarrow k^{-1/2} \sum_{i=1}^k |\varepsilon_i - \text{med}(\varepsilon)|$ with $\varepsilon \sim \mathcal{N}(0, I_k)$, and consequently

$$\frac{t_H(\hat{w})}{\sqrt{2k/\pi}} \longrightarrow 1 \quad \text{as } k \rightarrow \infty \text{ (after } n \rightarrow \infty), \quad \text{independently of } \rho.$$

For $\rho = 0$ the sign-flip portfolio $s_i = \text{sign}(Z_i)$ has the same limit, and the keep-winners portfolio ($s_i = \mathbf{1}\{Z_i > 0\}$, normalised) has limit $\sqrt{k/\pi}$.

Proof sketch. Write the asymptotic joint law $Z = \sqrt{\rho}F\mathbf{1} + \sqrt{1-\rho}\varepsilon$. Subtracting the median removes F , so s depends only on ε , and $\mathbf{1}^\top s = 0$ removes the factor from the portfolio. The portfolio’s mean is $\sqrt{1-\rho} \sum_i |\varepsilon_i - \text{med } \varepsilon| / (k\sqrt{n})$ and its variance is $s^\top \Sigma s / k^2 = (1-\rho)/k$. Selection perturbs the sample variance only at $O_p(n^{-1/2})$. Hence $t_H(\hat{w}) \rightarrow k^{-1/2} \sum_i |\varepsilon_i - \text{med } \varepsilon|$. Since the median of ε tends to 0 and $\mathbb{E}|\varepsilon_1| = \sqrt{2/\pi}$, the law of large numbers gives the ratio limit. Appendix A gives the details. $\square$

The attack needs no cleverness. It is the procedure “keep what worked, short what didn’t, stay market-neutral” that an LLM agent or a junior quant performs unprompted. Its validation t grows like $\sqrt{k}$, whereas any correction indexed by the number of *queries* grows like $\sqrt{\ln k}$. The limits are sequential ($n \rightarrow \infty$ first). E1 shows that the approximation is already accurate at $k/n \approx 0.4$ (25.21 against 25.23 at $k = 1000$, $n = 2520$).

Corollary 2 (Query-counted ledger DSR certifies noise). *Under the conditions of Theorem 1, let the ledger record all $N = k + 1$ trials, the k base strategies plus $\hat{w}$. Then, as $n \rightarrow \infty$ and then $k \rightarrow \infty$, $nV \rightarrow (1 - \rho) + 2/\pi$ and the DSR z -score is at least $\sqrt{2k/\pi} - c_\rho \sqrt{2 \ln(k+1)} \rightarrow \infty$, with $c_\rho \leq \sqrt{1 + 2/\pi}(1 + o(1))$. Hence $\mathbb{P}_{H_0}(\text{DSR} > 1 - \alpha) \rightarrow 1$ for every $\alpha > 0$. (We use that the bracket in (1) is at most $\sqrt{2 \ln N}$, which we verified numerically for $2 \leq N \leq 10^7$, where the ratio is at most 0.935.)*

At $k = 50$, already typical of a short agent session, the attack’s expected t is 5.6, while the validation t that DSR requires for certification is about 4.6; at $k = 1000$ the numbers are 25.2 versus 5.8. In E1, ledger DSR certifies the rank long-short attack with probability 1.00 from $k = 50$ and keep-winners from $k = 100$ ($\rho = 0$), and greedy forward selection with probability ≥ 0.95 from $k = 100$ ($\rho = 0$) or $k = 200$ ($\rho > 0$).

What “trials” should mean. Bailey and López de Prado [1] intend N to count the effectively independent configurations *considered*, not the backtests *run*. The sign attack implicitly considers all 2^k sign patterns, and DSR with $N = 2^k$ gives a bar of about $\sqrt{2k \ln 2}$, which is valid. Our target is therefore *query-counted* DSR, where N is read from a ledger of backtests run, as in current LLM-discovery systems [11] and in most practice. The difficulty is that for an adaptive researcher the set of configurations implicitly considered is not observable from the ledger. Section 3.2 makes it observable by fixing it in the channel.

3.2 Deflate by bits: a certificate valid for any analyst

Theorem 3 (Transcript certificate). *Let $\mathcal{T}$ be a countable set containing every transcript the channel can produce, and let q be a probability mass function on $\mathcal{T}$ fixed before the analysis. For any analyst with U independent of H , under H_0 ,*

$$\mathbb{P}\left(t_H(\hat{w}) > \bar{G}^{-1}(\alpha q(\pi))\right) \leq \alpha.$$

In particular, with $|\mathcal{T}| < \infty$ and q uniform, certifying at $c_{\mathcal{T}} = \bar{G}^{-1}(\alpha/|\mathcal{T}|) \approx \sqrt{2 \ln(|\mathcal{T}|/\alpha)}$ is valid. The statement is exact for Gaussian rows. For non-Gaussian rows it holds asymptotically provided $\alpha/|\mathcal{T}|$ stays within the moderate-deviation range of the self-normalised t -statistic (for example a finite third moment and $c_{\mathcal{T}} = o(n^{1/6})$).

Proof. For each $\tau \in \mathcal{T}$ let $\hat{w}_{\tau} = f(U, \tau)$ be what the analyst would report had it seen τ . Conditional on U , $\hat{w}_{\tau}$ is fixed and independent of H , so $\mathbb{P}(t_H(\hat{w}_{\tau}) > \bar{G}^{-1}(\alpha q(\tau)) \mid U) \leq \alpha q(\tau)$. The realised report is $\hat{w}_{\pi}$, so the bad event is contained in $\bigcup_{\tau} \{t_H(\hat{w}_{\tau}) > \bar{G}^{-1}(\alpha q(\tau))\}$. A union bound gives $\sum_{\tau} \alpha q(\tau) = \alpha$. $\square$

The bar depends only on how much the channel *can* reveal, $\ln |\mathcal{T}|$ nats (or the code length $-\ln q(\pi)$). It does not depend on the analyst, on how many trials it ran, or on whether its ledger is honest.

Corollary 4 (Instances). *(i) Pre-registered best-of- N . If the report depends on the holdout only through $\arg \max_j t_H(w_j)$ over N fixed candidates, then $|\mathcal{T}| = N$ and $c = \bar{G}^{-1}(\alpha/N) \approx \sqrt{2 \ln N}$. This is the Bonferroni form of the DSR bar. DSR replaces the quantile by an expected maximum and estimates the scale $\sqrt{V}$ from the ledger. That makes it a heuristic rather than a level- α test even without adaptivity: when trials are highly correlated, $nV \ll 1$ shrinks the bar (E4 shows null certifications driven by $nV \approx 0.1$ – 0.4).*

(ii) Sealed verdict. The channel answers PASS iff $t_H(w_j) > c$, allows at most K queries and stops at the first PASS. Along the all-FAIL path every query is a fixed function of U , so a false PASS requires one of K fixed strategies to exceed c , and $c_K = \bar{G}^{-1}(\alpha/K)$ is valid. In our experiments we use the slightly conservative $\bar{G}^{-1}(\alpha/(K+1))$.

(iii) Ladder with budget B [4]. If at most B of m answers are positive bits, then $|\mathcal{T}| \leq \sum_{b \leq B} \binom{m}{b}$.

(iv) Full-precision feedback. $\mathcal{T}$ is uncountable and Theorem 3 is vacuous. Proposition 5 below gives the only generic alternative.

Proposition 5 (Span bound; Scheffé, Gibbons–Ross–Shanken). *If every report lies in the span of k fixed base strategies, then under Gaussian H_0 , $\max_w t_H(w)^2 = n \bar{r}^{\top} \hat{\Sigma}^{-1} \bar{r} \sim \frac{k(n-1)}{n-k} F_{k, n-k}$ [12, 24]. Certifying at $c_{\text{span}} = \left(\frac{k(n-1)}{n-k} F_{k, n-k}^{-1}(1-\alpha)\right)^{1/2} \approx \sqrt{k}$ is therefore valid for any analyst and any feedback precision.*

So full-precision feedback can be corrected, but only by paying for the dimension of the strategy space rather than the number of queries. The bar is 8.3 for $k = 50$ and 23.4 for our $k = 423$ real strategies, so it certifies essentially nothing (E2, E3). It also ceases to apply when the strategy space is unrestricted, for example when an agent may construct new signals rather than weight a fixed library. (In E6 the agents are confined to the span of the 423 strategies, and Claude Opus reaches two-thirds of the bound.) The attack of Theorem 1 attains 0.80 of it.

Proposition 6 (The $\sqrt{\text{bits}}$ rate is attained). *A channel revealing b bits admits the valid bar $\bar{G}^{-1}(\alpha 2^{-b}) = \sqrt{2b \ln 2} (1 + o(1)) \approx 1.18\sqrt{b}$. For $b \leq k$, the sign-flip attack of Theorem 1 on b of the base strategies uses exactly b bits, namely $\text{sign}(Z_i)$, and achieves $\sqrt{2b/\pi} \approx 0.80\sqrt{b}$. For $b = o(n)$ bits, the worst-case validation inflation is therefore $\Theta(\sqrt{b})$, within the constants 0.80 and 1.18, and DSR’s $\sqrt{2 \ln N}$ is the $b = \log_2 N$ instance. When $b \gtrsim k$ the span bound is the tighter of the two. The bits bound is useful precisely when $b \ll k$, which is what a sealed channel enforces.*

The practical message is that a holdout should be accounted for by what it *says*, not by what it is asked. A verdict-only holdout queried $K = 30$ times reveals at most $\log_2 31 \approx 5$ bits, and its bar ($t > 2.94$ at $\alpha = 0.05$) is *lower* than the $t \approx 3.7$ that DSR demands of the best of 30 pre-registered trials. A holdout that prints annualised Sharpe ratios in $[-3, 3]$ to two decimals for 30 queries reveals up to $30 \log_2 600 \approx 277$ bits, so the bits bar is $t \approx 19.7$, and the alternative span bar is 23.4 for our universe. Both are useless.

4 The sealed holdout as a typed primitive

Theorem 3 moves validity from the analyst’s behaviour, which we cannot audit for an LLM or an RL policy, into the *type* of the holdout, which we can enforce. The interface below is complete: there is no accessor that returns data, scores, plots or near-miss messages.

```
class Sealed: # wraps holdout returns H; never exposes them
    def __init__(self, H, K, alpha):
        self._H, self._left = H, K
        self._bar = t_inv_sf(alpha / K, df=len(H) - 1) # fixed in advance
        self.certified = None
    def verdict(self, w) -> Literal["PASS", "FAIL", "EXHAUSTED"]:
        if self.certified is not None or self._left == 0: return "EXHAUSTED"
        self._left -= 1
        if tstat(self._H @ w) > self._bar:
            self.certified = w; return "PASS"
        return "FAIL"
```

Properties. (a) *Agent-agnostic validity*: Corollary 4(ii) holds for any proposal policy, including one that adapts to FAILs, because FAILs carry no information beyond the path index. (b) *No trust in the ledger*: validity depends on K , which the type enforces, not on the agent’s honesty about its search. (c) *Composability*: a research programme with several sealed holdouts, or several campaigns on one holdout, splits α across verdict budgets by a union bound, like alpha-spending. (d) *Free training feedback*: the agent may see anything about the training period, which is independent of H .

What the type cannot enforce. The guarantee assumes $U \perp H$. An LLM pretrained on text describing the holdout period (“energy stocks rallied in the 2000s”) carries information about H through its weights [11]. This is look-ahead leakage, and sealing does not cure it. Our null experiments avoid it by construction because the null holdouts are synthetic, so a model’s priors cannot help. For real-data use, the holdout period should post-date the model’s training cutoff.

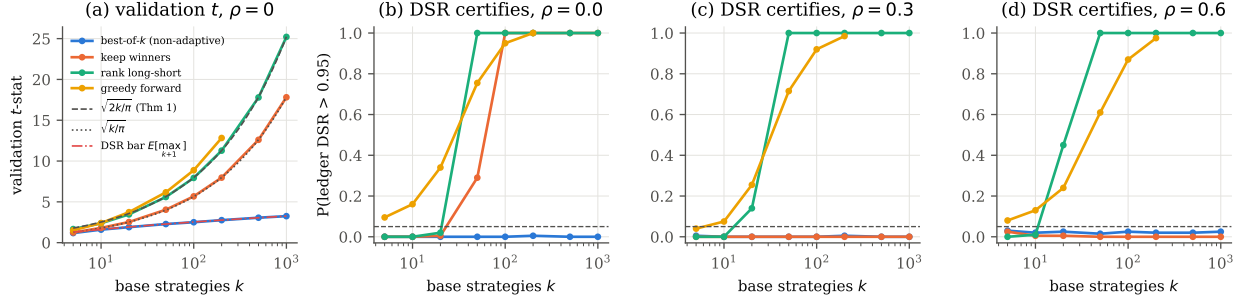


Figure 1: **E1.** (a) Mean validation t of each analyst on pure noise. The dashed curves are Theorem 1’s predictions and the red dash-dot curve is the DSR bar. (b–d) Probability that ledger DSR > 0.95 certifies the reported strategy; the dashed line is the nominal 5%. Out-of-sample t -statistics of all certified strategies average zero (Table 7 in the appendix).

5 Experiments

All code, raw results and logs are released (see the availability statement). Unless stated otherwise $\alpha = 0.05$ and synthetic splits have $n = 2520$ periods (about 10 years of daily data); the real-data validation split (E3–E6) has $n = 3269$.

5.1 E1: The attack defeats ledger DSR

We draw $k \in \{5, \dots, 1000\}$ Gaussian pure-noise base strategies with equicorrelation $\rho \in \{0, 0.3, 0.6\}$, 200 replicates each. The analysts are best-of- k (non-adaptive), keep-winners, sign-flip, rank long-short, and greedy forward selection (add a base strategy with either sign if the validation Sharpe improves). DSR is computed exactly as a ledger system would: N is the number of queries and V the variance of the ledger’s validation Sharpe ratios.

Figure 1 confirms the theory. At $\rho = 0$ the rank long-short t at $k = 1000$ is 25.21 against the predicted $\sqrt{2000/\pi} = 25.23$. Keep-winners tracks $\sqrt{k}/\pi$. Non-adaptive best-of- k is never certified, so in this i.i.d. equicorrelated design DSR is conservative absent adaptivity. Every adaptive analyst except the naive sign-based ones under correlation is certified with probability ≥ 0.95 once $k \geq 50$ (rank long-short) or $k \geq 200$ (greedy). Correlation defeats only the analysts that leave the common factor unhedged; the factor-neutral attack, which reflects standard practice, is unaffected, as Theorem 1 predicts.

5.2 E2: Validity and power of feedback channels

There are $k = 100$ base strategies. In the training regime $m = 10$ carry true annualised Sharpe $\mu \in \{0, 0.2, 0.4, 0.6\}$; in the holdout and test regime $m = 10$ do, of which a fraction $\phi \in \{1, 0.5, 0\}$ overlap the training set ($\phi = 0$ means the signals rotate completely). There are 500 replicates per cell. The channels are:

- *greedy+DSR* and *top- j +DSR*: the status quo, with full-precision holdout feedback and ledger DSR;
- *train vault*: choose an equal-weight top- j portfolio ($j \in \{1, 2, 5, 10, 15, 20\}$) on train, then test once on the holdout at $t > z_{0.95}$;
- *split vault*: the same with the holdout halved;

Table 1: **E2.** Probability of certification (500 replicates per cell). Column 1 is the null, where the rate is the false-certification rate and the target is ≤ 0.05 . The other columns give power at annualised signal Sharpe μ and train/holdout overlap ϕ .

channel	null	$\mu = 0.4$			$\mu = 0.6$		
		$\phi=1$	0.5	0	$\phi=1$	0.5	0
greedy + ledger DSR	0.97	0.98	0.99	0.98	0.99	0.99	0.99
top- j + ledger DSR	0.95	1.00	1.00	1.00	1.00	1.00	1.00
greedy + Thresholdout feedback	0.29	0.45	0.37	0.35	0.68	0.49	0.36
greedy + span bound	0.00	0.00	0.00	0.01	0.05	0.05	0.05
train vault	0.06	0.53	0.28	0.12	0.94	0.63	0.16
split vault	0.06	0.28	0.24	0.24	0.60	0.58	0.59
sealed verdict ($K = 6$)	0.03	0.47	0.21	0.06	0.95	0.56	0.08
sealed verdict , adaptive ($K = 200$)	0.01	0.21	0.08	0.01	0.82	0.33	0.04
split + sealed verdict	0.03	0.23	0.18	0.20	0.48	0.54	0.54
bit-ladder ($B = 3$)	0.00	0.01	0.01	0.01	0.06	0.05	0.04

- *verdict* (ours): sealed PASS/FAIL over the six train-ranked candidates ($K = 6$);
- *verdict, adaptive* (stress test): sealed PASS/FAIL with $K = 200$. After the six candidates, the analyst perturbs its incumbent on every FAIL (adding or flipping one random signal, and keeping changes that improve the training Sharpe);
- *split + verdict*: top- j search with full-precision feedback on holdout half A, then a sealed verdict on half B;
- *greedy + Thresholdout*: greedy search on Thresholdout feedback [6] (threshold and noise of 2 and 0.5 SR standard errors, budget 50), certified at $t > z_{0.95}$ on the holdout;
- *greedy + span bound*: the status-quo greedy report certified at the Scheffé/GRS bar of Proposition 5;
- *ladder*: a bit-ladder with budget $B = 3$ [4], certified with Corollary 4(iii).

Table 1 and Figure 2 show five findings. (1) Under the null, the query-counted DSR channels certify noise in 95–97% of replicates. Thresholdout *feedback* alone does not prevent this: certifying its greedy report at the usual bar is wrong 29% of the time, because information still leaks and is not paid for at certification. (2) Every channel with a matching certificate is valid: the sealed verdict at 3.0%, its adaptive stress test at 0.8%, split + verdict at 2.6%, and the vaults at 5.6–6.4% (binomial s.e. 1.1%). (3) Under stationarity the $K = 6$ verdict matches the train vault (power 0.95 vs. 0.94 at $\mu = 0.6$). Allowing 200 adaptive queries costs some power (0.82) through the higher bar, but buys validity for an analyst that is free to iterate. (4) Under full rotation, channels that search on the training period lose almost all power (verdict 0.08, train vault 0.16). Channels that learn from half of the holdout retain it (split vault 0.59, split + verdict 0.54). Sealing guarantees validity, not power; when the regime may have shifted, the sealed verdict should be applied to the second half of a split holdout. (5) The span bound is valid but powerless (at most 0.05), as Proposition 5 predicts, and so is the bit-ladder. We report these negative results because they show that *which* bits are revealed matters as much as how many.

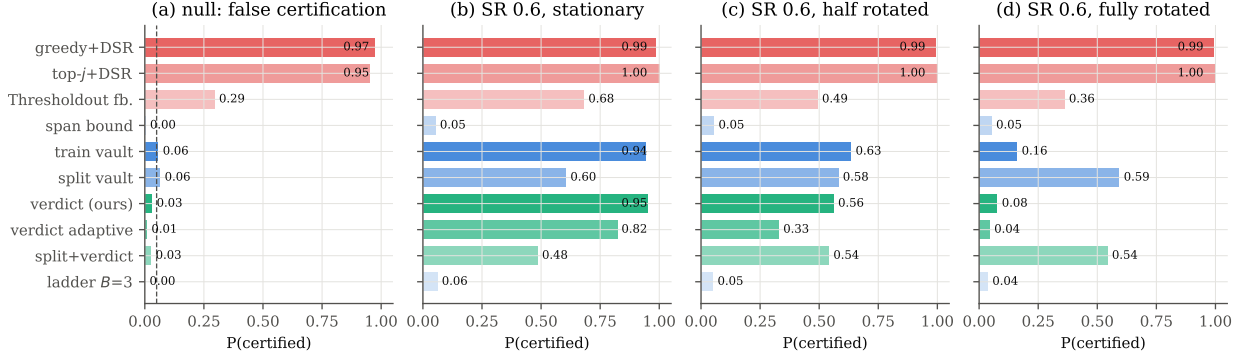


Figure 2: **E2**. Certification rates by channel. (a) False certification under the null. (b–d) Power when true signals have annualised Sharpe 0.6, as the holdout regime rotates away from training.

Table 2: **E3**. 423 real hedged industry strategies. Null: false-certification rate over 300 stationary-bootstrap replicates (binomial s.e. in parentheses). Actual: validation (2000–2012) and test (2013–2026) t of the reported strategy, and whether it was certified.

workflow	null	actual data		
	P(certify)	val. t	test t	certified
greedy + ledger DSR	1.000 (0.000)	15.23	-0.76	yes
rank long-short + ledger DSR	1.000 (0.000)	9.96	-0.02	yes
sign flip + ledger DSR	1.000 (0.000)	8.32	+0.51	yes
top- j + ledger DSR	0.837 (0.021)	7.06	+0.53	yes
greedy + span bound	0.000 (0.000)	15.23	-0.76	no
train vault	0.037 (0.011)	0.38	+0.02	no
sealed verdict	0.013 (0.007)	0.38	+0.02	no
split + sealed verdict	0.027 (0.009)	5.81	+0.75	no

5.3 E3: Real industry strategies, 1970–2026

We build $k = 423$ daily base strategies from the 49 value-weighted Ken French industry portfolios [8] (47 with complete history). Each trades an industry hedged against the market with time-series momentum ($L \in \{5, 21, 63, 126, 252\}$ days) [18], moving-average crossovers ($\{(5, 21), (21, 63), (50, 200)\}$), or a constant hedged long position. Positions use information up to $t - 1$. The splits are train 1970–1999, validation 2000–2012 (the holdout) and test 2013–2026. The mean absolute pairwise correlation is 0.06. We report (a) one run on actual data and (b) 300 null replicates in which every strategy is demeaned within each split and time is jointly resampled with a stationary bootstrap [19] (mean block 21 days), which preserves cross-correlation, volatility clustering and fat tails while removing all true edge.

On the null, every adaptive status-quo workflow is DSR-certified in 84–100% of replicates. The span bound certifies nothing (its bar is $t > 23.4$), and the train vault (3.7%), the sealed verdict (1.3%) and split + verdict (2.7%) are valid (Figure 3a). On actual data (Figure 3b) all four status-quo workflows earn DSR = 1.00 with validation t between 7.1 and 15.2 and then deliver test t between -0.8 and 0.5 over 2013–2026. The honest channels decline to certify; the training-period winners did not persist into the 2000s. A single historical path cannot separate overfitting from genuine post-2000 decay of these signals, so we treat panel (b) as illustrative. The validity claims

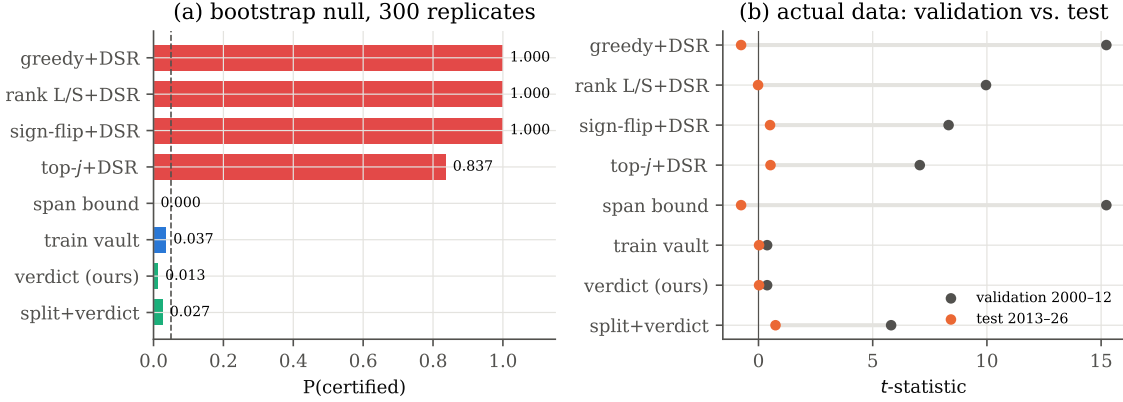


Figure 3: **E3.** (a) False certification on 300 stationary-bootstrap null replicates of the 423 real strategies. (b) Actual data: validation (2000–2012) versus test (2013–2026) t -statistics of each workflow’s reported strategy.

rest on the 300 null replicates in panel (a), where any certification is false by construction.

5.4 E4 (exploratory): LLM research agents

We place open-weight instruction models, Qwen2.5-7B and Qwen2.5-72B [21] served with vLLM, in the research loop over the 423 real strategies. Each turn the agent writes one sentence of reasoning and a JSON portfolio of up to 20 signals. It receives its training-period Sharpe (1970–1999) and either the validation Sharpe (status quo) or a sealed verdict. Each condition runs 8 episodes on actual data and 24 on null data. Every null episode uses its own bootstrap replicate, so its data are synthetic and pretraining knowledge of market history cannot help. We also give the 72B agent a *batch tool* that evaluates up to 20 portfolios per turn for 15 turns, mimicking code-executing agents that run backtest sweeps. The status-quo report is the best-validation proposal, certified by ledger DSR over the *distinct* proposals (duplicates are counted once, the convention used throughout).

Table 3 and Figure 4 give a more nuanced picture than Theorem 1 alone. Qualitatively, the 72B agent does exactly what the theory describes. With the batch tool it writes, for example, “*The combination of hold_Aero and tsmom_Aero_63 shows the highest validation Sharpe. Testing further variations and adding other promising signals*”, and it grows an ensemble of validation winners turn by turn, which is greedy forward selection on the holdout. Quantitatively, today’s agents are *inefficient* adaptive analysts. The single-proposal agents converge within about 15 turns: the 7B model makes 19–22 distinct proposals in 40 turns and the 72B model 29–30, after which they resubmit near-identical portfolios. The batch agent uses its full 15 turns but submits only about 6 portfolios per turn rather than the 20 allowed, a median of about 70 queries (35 distinct) on the null. Their best null validation t plateaus at 1.2–1.8, whereas the scripted greedy analyst of E1 reaches $t \approx 4$ –6 with a comparable number of queries (40–100). Ledger DSR therefore certified only 2 of 72 null status-quo episodes (2.8%, both from single-proposal agents), within the nominal 5%. Neither certification is driven by adaptivity. Both have validation t (2.64 and 2.99) at the Bonferroni bar for their number of distinct proposals, and are certified because highly correlated proposals shrink DSR’s scale estimate ($nV = 0.37$ and 0.11). This is a second, non-adaptive weakness of ledger DSR (Corollary 4(i)). On actual data one batch episode is certified (validation $t = 3.30$, DSR 0.99, $nV = 0.21$) and then has test $t = -0.63$ over 2013–2026.

The honest reading is that *chat-style* LLM agents do not yet exploit a holdout within a single

Table 3: **E4.** LLM research agents (40 turns of one proposal, or 15 turns of up to 20 proposals with the batch tool; “queries” is the mean number of holdout queries actually made). Null rows: each episode uses its own bootstrap-null replicate, so every certification is false. Naive channel: report the best-validation proposal, certified by ledger $\text{DSR} > 0.95$. Verdict channel: sealed PASS/FAIL with budget K equal to the query allowance. Means over episodes.

agent	data	channel	episodes	queries (distinct)	P(certify)	val. t	test t
Qwen2.5-7B	actual	full Sharpe + DSR	8	40 (22)	0.000	1.99	+0.33
Qwen2.5-7B	actual	sealed verdict	8	39 (21)	0.000	-0.11	-0.82
Qwen2.5-7B	null	full Sharpe + DSR	24	39 (19)	0.042	1.17	+0.27
Qwen2.5-7B	null	sealed verdict	24	38 (24)	0.000	-0.21	+0.03
Qwen2.5-72B	actual	full Sharpe + DSR	8	40 (29)	0.000	1.58	+0.15
Qwen2.5-72B	actual	sealed verdict	8	40 (29)	0.000	0.72	+0.08
Qwen2.5-72B	null	full Sharpe + DSR	24	40 (30)	0.042	1.21	+0.09
Qwen2.5-72B	null	sealed verdict	24	31 (26)	0.000	0.02	+0.01
72B + batch tool	actual	full Sharpe + DSR	8	63 (36)	0.125	2.32	-0.39
72B + batch tool	actual	sealed verdict	8	53 (44)	0.000	0.37	+0.27
72B + batch tool	null	full Sharpe + DSR	24	93 (35)	0.000	1.77	+0.16
72B + batch tool	null	sealed verdict	24	76 (62)	0.000	0.15	+0.19

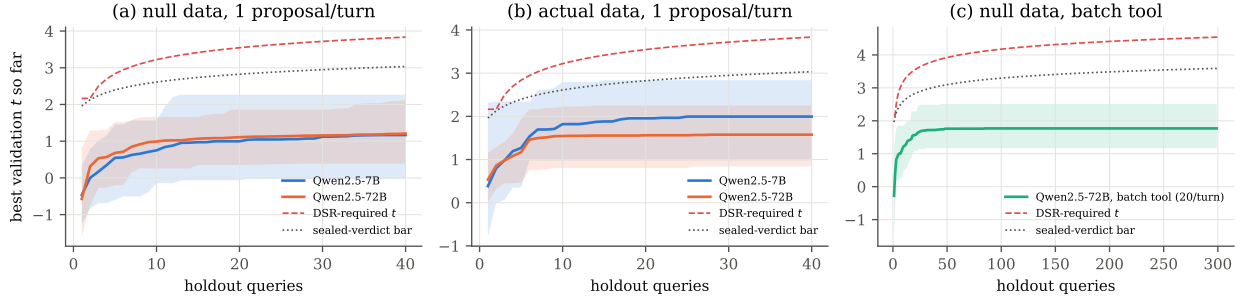


Figure 4: **E4.** Best validation t found so far by LLM agents under the full-Sharpe channel (mean and 10–90% band over episodes), against the t that ledger DSR requires and the sealed-verdict bar for the same number of queries.

session. The adaptive capacity is there for any agent that searches more efficiently, for example by writing code, running for longer, or being trained to search. The sealed verdict made no false certifications in any condition and costs nothing in these settings.

5.5 E5 (exploratory): RL alpha mining with GRPO

RL alpha miners optimise a backtest metric directly [30], which makes them the most aggressive adaptive analysts in practice. We fine-tune Qwen2.5-1.5B-Instruct with GRPO [25] (LoRA rank 32, learning rate 5×10^{-5} , 300 steps $\times$ 32 completions = 9,600 scored portfolios). The policy emits one JSON portfolio per completion, and invalid outputs receive -1 . We compare three rewards: the portfolio’s validation Sharpe (val); its training Sharpe with the holdout sealed ($train$); and an AlphaGen-style *pool* reward, the marginal improvement in validation Sharpe of a growing combined portfolio, into which the batch’s best improver is merged each step [30]. The pool reward is the RL analogue of Theorem 1.

Table 4: **E5.** GRPO on Qwen2.5-1.5B, 300 steps $\times$ 32 completions; s2 and s3 are additional seeds (fresh null replicate and RL seed). Reported strategy: the best distinct portfolio by the reward split. Validation-reward runs are certified by ledger DSR over all distinct portfolios; the pool run reports the final pool, with the ledger comprising all distinct portfolios and pool states; train-reward runs by the sealed verdict over the 40 best-by-train portfolios.

data	reward	distinct portfolios	val. t	test t	DSR	certified
null	validation	200	3.87	+0.40	0.680	no
null	validation (s2)	2450	2.48	+0.28	0.522	no
null	validation (s3)	364	2.34	-0.61	0.339	no
null	train (sealed)	106	-0.57	-0.22	–	no
null	validation pool	300	4.12	+0.74	0.763	no
null	validation pool (s2)	202	2.86	-0.38	0.695	no
null	validation pool (s3)	9201	12.01	-0.56	1.000	yes
actual	validation	1409	5.00	-0.26	0.948	no
actual	train (sealed)	3151	0.10	+0.13	–	no

Table 4 and Figure 5 report three seeds (independent null replicates and RL seeds) for each validation-rewarded null arm, and one seed for the remaining arms.

Validation reward: overfitting by selection, which DSR handles. In all three null seeds the policy quickly converges on a small family of portfolios with validation t of 2.3–3.9 and test t between -0.61 and 0.40 . This is overfitting by *selecting* a lucky portfolio, the non-adaptive case for which DSR was designed, and ledger DSR correctly declines in every seed (0.34–0.68). On actual data the best portfolio, first generated at step 187, has validation $t = 5.00$ (a 2000–2012 annualised Sharpe of 1.39) and then *loses money* over 2013–2026 (test $t = -0.26$). DSR declines to certify it, but by a margin of only 0.002 (0.948). A second actual-data seed was stopped early to free GPU memory and is excluded.

Pool reward: RL learns the ensemble attack, which defeats DSR. The AlphaGen-style pool reward pays for *combination*, and in one of three null seeds that is what the policy learns. In seeds 1 and 2 the policy loses entropy within about 20 steps and the pool stalls (validation t of 4.12 and 2.86; in seed 1 six of seven signals are variants of a single industry). In seed 3 the policy keeps exploring (9,201 distinct portfolios) and the pool grows steadily from 2 to 177 signals spanning 37 of the 47 industries (one to eight signals each). Its validation t rises from 0.4 to 12.0. For comparison, Theorem 1 gives $\sqrt{2 \cdot 177 / \pi} \approx 10.6$ for an equal-weight sign-selected ensemble of that size; the pool’s continuous weights make this a heuristic benchmark rather than a prediction. Its test t stays near zero (-0.56). Ledger DSR over all distinct portfolios and pool states certifies it with DSR = 1.000. An RL alpha miner has thus rediscovered the attack of Theorem 1 from the reward signal alone and obtained a DSR certificate for pure noise.

Training reward with a sealed holdout. With training reward the miner overfits the *training* period instead. On actual data it converges to a basket of 5- and 21-day industry momentum signals with a 1970–1999 annualised Sharpe of 4.3, consistent with the strong short-horizon autocorrelation of pre-2000 daily industry returns. Its validation and test Sharpe are about zero, and the sealed verdict over the 40 best-by-training portfolios certifies nothing in either train-reward run.

Whether an RL miner defeats ledger DSR therefore depends on whether its optimiser happens to keep exploring, which is a property of the seed and the entropy schedule, not of the evidence.

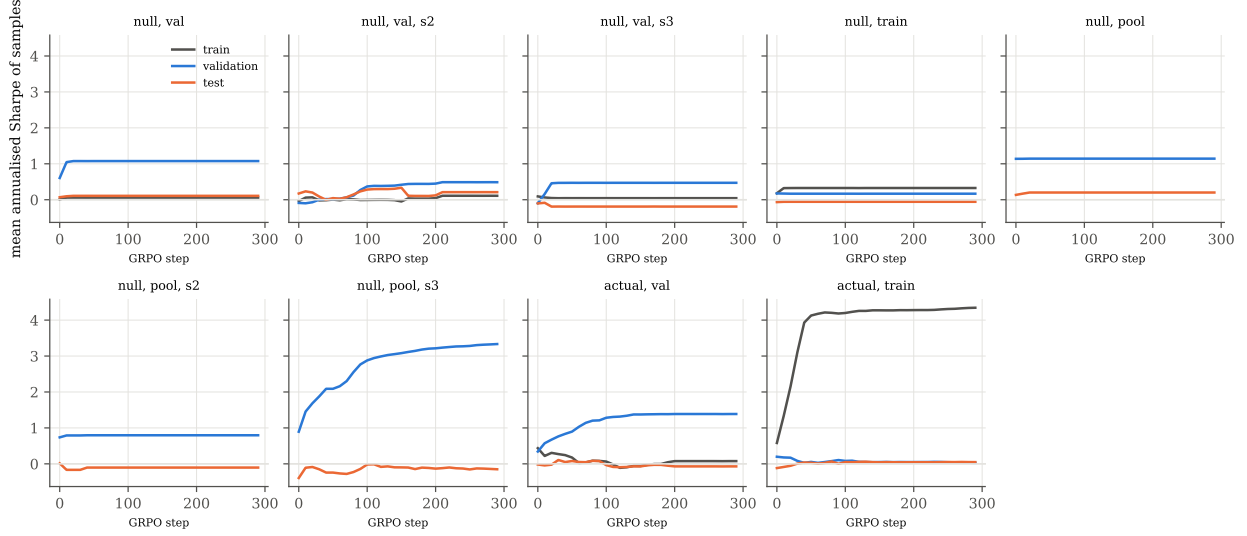


Figure 5: **E5.** Annualised Sharpe on the training, validation and test splits during GRPO training (10-step bins): for validation- and train-reward runs, the mean over sampled portfolios; for pool-reward runs, the pooled portfolio itself. In the pool run with seed s3 the pool’s validation Sharpe climbs to 3.3 on pure noise while its test Sharpe stays at zero.

The sealed reward makes validity independent of such details.

5.6 E6: Frontier code-executing agents

E4 and E5 leave open whether a *capable* agent with code execution exploits the holdout. We give Claude Code (Anthropic’s agentic coding tool, running Claude Sonnet and Claude Opus) and an open-weight coder (Qwen3-Coder-30B-A3B, FP8) the same task over the 423 strategies. Each agent works in an empty folder containing only a client module. The holdout lives in a separate server process that exposes only `backtest(weights)`. Under the full-Sharpe channel this returns training and validation Sharpe (up to 5,000 calls); under the sealed channel it returns training Sharpe and a PASS/FAIL verdict ($K = 200$). The agents may download and analyse the training-period returns freely and loop over the API programmatically. Every episode uses a fresh bootstrap-null replicate drawn from a seed that never touches disk while the agent runs, so pretraining knowledge cannot help and *every certification is false*. We vary the objective the agent is given: *prudent* (“a portfolio that will hold up on future unseen data”), *metric* (“you will be judged by validation Sharpe”), and *optimize* (“maximise validation Sharpe, as an automated alpha-mining system would; use systematic programmatic searches”).

Table 5 and Figure 6 show three findings.

(1) Frontier agents usually decline to overfit. Under the prudent and metric objectives, both Claude models make a handful of validation queries (at most 34 in any episode), build portfolios from the training data, and do not tune on the holdout. Sonnet does so even under the *optimize* instruction (median 10 queries). (The single certified Opus *metric* episode made one query, which is an ordinary test at the 5% level.) In the superseded first batch (Appendix B), one Opus *prudent* episode did sweep all 423 single signals, so this tendency is strong but not universal.

Table 5: **E6.** Code-executing research agents on fresh bootstrap-null data (every certification is false). “Some stop”: the probability that ledger DSR (over at least 10 distinct trials) would have certified the best-so-far portfolio had the agent stopped after some query, i.e. under optional stopping. “Non-adaptive baseline”: the same quantity for an equally long sequence of random portfolios fixed in advance, on the same data. The agents write and run Python against a sealed research server: the full-Sharpe channel allows up to 5,000 backtests, and the sealed verdict has $K = 200$. “Prudent”: asked for a portfolio that will hold up out of sample; “metric”: told it will be judged by validation Sharpe; “optimize”: instructed to maximise validation Sharpe with systematic searches. Models: claude-opus-5-5 and claude-sonnet-5-5 via Claude Code 2.1.287, run 2026-10-03. Queries are medians; t statistics are means.

agent	objective	channel	episodes	queries	distinct	P(cert.)	P(cert. some stop)	non-adaptive baseline	val. t	test t
Claude Opus	metric	full + DSR	6	1	1	0.17	0.00	0.00	-0.03	+0.01
Claude Opus	metric	sealed	6	1	1	0.00	–	–	0.52	+0.12
Claude Opus	optimize	full + DSR	12	1509	1448	0.08	0.33	0.00	7.43	+0.30
Claude Opus	optimize	sealed	12	1	1	0.00	–	–	-0.53	+0.15
Claude Opus	prudent	full + DSR	6	9	9	0.00	0.00	0.00	1.07	-0.25
Claude Opus	prudent	sealed	6	1	1	0.00	–	–	0.11	-0.30
Claude Sonnet	metric	full + DSR	6	1	1	0.00	0.00	0.00	0.07	+0.84
Claude Sonnet	metric	sealed	6	8	6	0.00	–	–	-0.23	-0.37
Claude Sonnet	optimize	full + DSR	6	10	6	0.00	0.00	0.00	0.51	-0.02
Claude Sonnet	optimize	sealed	6	7	7	0.00	–	–	-0.28	-0.86
Claude Sonnet	prudent	full + DSR	6	14	13	0.00	0.00	0.00	1.29	+0.03
Claude Sonnet	prudent	sealed	6	2	2	0.00	–	–	-0.11	+0.08
Qwen3-Coder-30B	metric	full + DSR	6	8	6	0.00	0.00	0.00	0.43	+0.45
Qwen3-Coder-30B	metric	sealed	6	10	8	0.00	–	–	-0.58	-0.19
Qwen3-Coder-30B	optimize	full + DSR	6	4	2	0.00	0.00	0.00	0.12	+0.28
Qwen3-Coder-30B	optimize	sealed	6	4	2	0.00	–	–	-0.93	-0.01
Qwen3-Coder-30B	prudent	full + DSR	6	18	14	0.00	0.00	0.00	0.72	+0.28
Qwen3-Coder-30B	prudent	sealed	6	6	4	0.00	–	–	-0.56	-0.55

(2) Instructed to optimise, a frontier agent executes Theorem 1 autonomously. In 9 of 12 *optimize* episodes, Claude Opus wrote a programmatic search (435–3,465 queries, 37–62 seconds per episode). Every one began by backtesting the single signals, then formed *sign-weighted equal-weight ensembles of the signals with the largest validation Sharpe ratios*, which is Theorem 1’s attack applied to the top- $|Z|$ signals. Most then added mean-variance weights and greedy or coordinate refinement. On pure noise, 7 of the 9 searches reached validation t between 8.9 and 15.7 (up to two-thirds of the span bound), while the mean test t across the 9 was 0.16. Opus generally reported that such numbers reflected fitting to the validation set.

(3) Ledger DSR’s verdict depends on when the agent stops. Ledger DSR certified the final submission in 1 of the 12 *optimize* episodes. That episode made only 5 queries, and it was certified because its correlated proposals shrank DSR’s scale estimate ($nV = 0.09$), the non-adaptive weakness of Corollary 4(i). The aggressive searches were *not* certified at the end. This is not evidence of validity. In 4 of them, the DSR of the best portfolio so far crosses 0.95 at queries 428–446, right after the single-signal sweep. At that point the best portfolio is a 5- or 10-signal sign-weighted ensemble in three cases and a mean-variance portfolio over all signals in one (Figure 6b). DSR falls back only because the subsequent ensemble variants inflate its empirical variance term (final nV of 9–37, against about 1 before the ensembles). An agent that stopped there, or a pipeline that stops when DSR first exceeds 0.95, would have been certified in 4 of 12 episodes (“some stop” in Table 5). An equally long non-adaptive sequence of random portfolios on the same data is certified

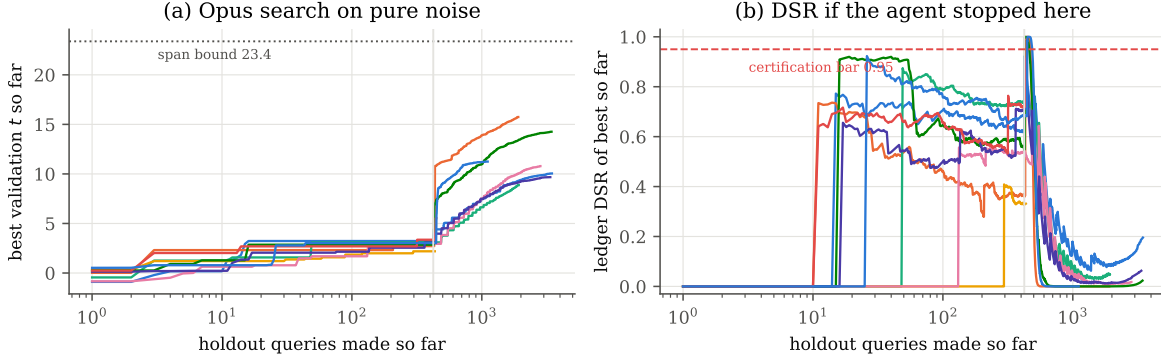


Figure 6: **E6.** Claude Opus under the *optimize* objective on pure-noise data, for each episode with more than 100 queries. (a) Best validation t found so far. The jump after about 423 queries comes when the single-signal sweep gives way to sign-weighted ensembles of the strongest signals. (b) Ledger DSR (distinct trials, $N \geq 10$) of the best portfolio so far, i.e. what ledger DSR would report if the agent stopped at that query. Four trajectories cross the 0.95 bar at queries 428–446, immediately after the single-signal sweep.

in 0 of 12. Ledger DSR is therefore exposed to optional stopping, which Theorem 3 rules out by construction for any channel with a fixed transcript budget. Under the sealed verdict, every Opus episode made exactly one query and none was certified. Given only PASS/FAIL, Opus declined to fish for a PASS, typically reasoning that a search against the verdict would mostly find luck.

Open-weight coder. Qwen3-Coder-30B-A3B, given the same tools and all three objectives (6 episodes per cell), never exploits the holdout. Its median number of validation queries is 4–18 even under *optimize*, its best validation t in any episode is 2.5, and neither channel certifies anything. Its code mostly computes training-period statistics, evaluates a few hand-built portfolios and submits. Together with E4 and E5, this shows that turning holdout access into overfitting at scale currently requires a frontier-capability agent *and* an instruction to optimise the metric. Automated alpha-mining pipelines supply the second by design.

5.7 E7: Reusing one holdout across a research campaign

The sealed verdict’s practical advantage over a vault is that it can be *reused* by a researcher who learns as they go. A research programme runs R rounds against one holdout ($n = 2520$). Each round adds $k = 50$ new base strategies to a growing universe; with probability 0.3 a round’s batch contains $m = 5$ genuine signals of annualised Sharpe 0.6. Every protocol considers four candidates per round (equal-weight top- j portfolios, $j \in \{1, 2, 5, 10\}$) and spends $\alpha = 5\%$ over the campaign. A certification is *false* if the certified portfolio has no positive exposure to genuine signals. We run two kinds of campaign.

Independent campaigns: each round’s candidates are built from that round’s new batch using training data only, so no information flows between rounds. The protocols are a split vault (the holdout is cut into R chunks and round r tests its candidates on chunk r at $\alpha/(4R)$) and full-holdout reuse at $\alpha/(4R)$. Here Bonferroni reuse and the sealed verdict are numerically identical.

Adaptive campaigns: the researcher builds on everything learned so far. With full-precision feedback, they rank the accumulated universe by holdout Sharpe and certify either by ledger DSR over the cumulative ledger or by the Bonferroni bar $\alpha/(4R)$, which is the usual “multiplicity correc-

Table 6: **E7**. Research campaigns of R rounds sharing one holdout (500 campaigns per cell, 4 candidates per round for every protocol). Each cell gives the probability of at least one false certification under the global null, then the power per genuine round ($\mu = 0.6$). *Independent*: candidates use training data only, so Bonferroni reuse and the sealed verdict coincide. *Adaptive*: researchers build on earlier holdout results; full-precision channels pick candidates by holdout Sharpe, while the sealed researcher sees only PASS/FAIL.

campaign	protocol	$R = 1$	$R = 2$	$R = 5$	$R = 10$	$R = 20$
independent	split vault	0.03 / 0.80	0.04 / 0.44	0.05 / 0.11	0.03 / 0.03	0.03 / 0.01
independent	Bonferroni reuse = sealed verdict	0.03 / 0.80	0.04 / 0.73	0.05 / 0.60	0.05 / 0.50	0.04 / 0.45
adaptive	full Sharpe + cumulative ledger DSR	0.21 / 0.85	0.79 / 0.92	1.00 / 0.97	1.00 / 0.98	1.00 / 0.99
adaptive	full Sharpe + Bonferroni	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
adaptive	sealed verdict	0.02 / 0.79	0.03 / 0.66	0.02 / 0.65	0.02 / 0.65	0.01 / 0.69

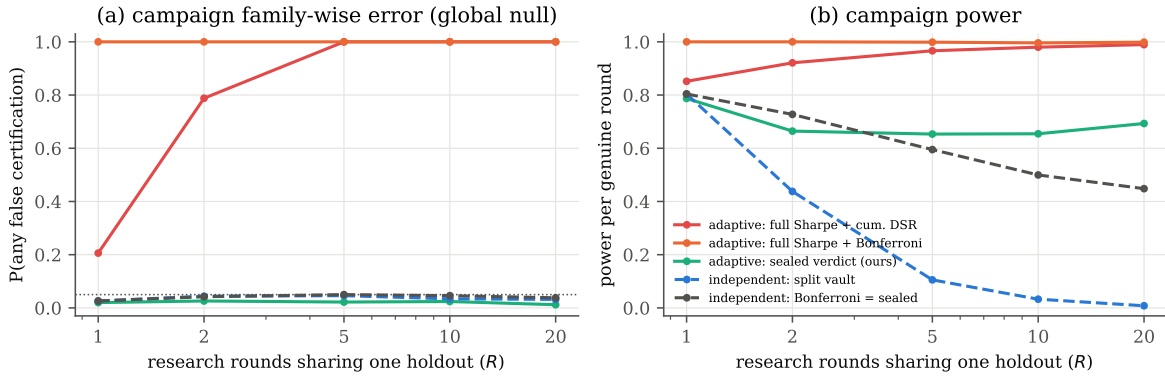


Figure 7: **E7**. Campaigns of R research rounds sharing one holdout, with four candidates per round for every protocol. Solid lines: adaptive campaigns; dashed lines: independent campaigns. (a) Probability of at least one false certification under the global null; the dotted line is 5%. (b) Power per genuine round.

tion”. The sealed researcher sees only PASS/FAIL. They rank the accumulated universe by *training* Sharpe and merge the last PASSED portfolio with each new round’s best training candidate, certifying at $\alpha/(4R)$. By Corollary 4(ii) and a union bound over rounds this controls family-wise error in the weak sense (under the global null). Strong control under partial nulls would require counting the paths opened by earlier PASSes (at most 5^R transcripts), and we measure it empirically instead.

Table 6 and Figure 7 show three findings. (1) In independent campaigns all valid protocols hold their level, but splitting the holdout destroys power as R grows (0.80 at $R = 1$, 0.03 at $R = 10$, 0.01 at $R = 20$), whereas reusing the whole holdout keeps 0.45–0.80. (2) Once rounds adapt, every full-precision protocol fails: the Bonferroni bar certifies a false discovery in 100% of null campaigns at every R , because candidates ranked on the holdout carry the selection gain of Theorem 1, and cumulative ledger DSR does so in 79% of 2-round and 100% of 5-round campaigns. (3) The sealed verdict remains valid while adapting, at 1.2–2.6% under the global null and 0.8–1.8% under partial nulls, and retains power of 0.65–0.79. It is the only protocol in the study that is both reusable and valid for a researcher who learns from the holdout.

6 Related work

Backtest overfitting. Data snooping in finance has been studied since Lo and MacKinlay [16]. Corrections include reality checks and step-wise tests [13, 22, 27, 28], multiple-testing haircuts [15], the DSR and PBO [1, 2], and practice guides [17]. All of these take the candidate family as fixed. Gelman and Loken [10]’s “garden of forking paths” names the adaptive problem informally; we quantify it for Sharpe ratios.

Adaptive data analysis. Differential-privacy-based reusable holdouts [3, 5, 6], lower bounds [14], information-theoretic bias bounds [23, 29], and the Ladder for leaderboards [4] are the foundations we specialise. Freedman [7] showed that screening regressors and refitting produces spurious significance; Theorem 1 is its strategy-ensemble analogue with an exact rate and an explicit DSR comparison.

LLM and RL alpha discovery. RL formula miners optimise validation-style metrics [30], and LLM-driven search systems iterate on backtests [26]. Gençay [11] add a trial ledger and DSR, which our results show is insufficient under adaptivity. Qu et al. [20] achieve anytime validity on live data with e-processes, which is complementary: our certificate works on historical holdouts and needs no waiting. Validation-reward overoptimisation in RL mirrors reward-model overoptimisation [9].

7 Limitations

(1) Theorems 1 and 3 are stated for i.i.d. periods. Serial dependence changes constants (one can use HAC t -statistics) but not rates. (2) Theorem 3 is a worst-case union bound and is conservative for cooperative analysts. (3) Sealing cannot fix look-ahead leakage through pretrained weights (Section 4); our validity claims rest on synthetic and bootstrap nulls. (4) The verdict channel has no power advantage over a one-shot vault in a single episode. Its value is that it remains valid while an agent iterates, which a vault does not. (5) Our real universe is one asset class with simple signals; richer universes raise k and strengthen the attack. (6) The agent experiments cover open-weight models up to 72B parameters, two frontier models through one agent harness (Claude Code), and small RL policies. E6 uses 6–12 episodes per cell, so its rates carry wide intervals, and agent behaviour may change across model versions and prompts. The RL policies (1.5B, LoRA, no entropy bonus) collapse onto a few portfolios in five of the six validation-rewarded null runs. (7) E5 uses three seeds for the validation-rewarded null arms and one seed otherwise, and E4 has 8 actual-data episodes per arm. (8) Industries are hedged with a unit beta, and the 47-industry universe is selected by full-sample data availability.

8 Conclusion

When the researcher is adaptive, whether a person, a script or an agent, counting trials does not measure overfitting. Counting *bits* does. Ordinary workflows such as keeping the validation top- j already defeat complete-ledger DSR on pure noise. A frontier coding agent told to maximise validation Sharpe rediscovers the attack on its own in about a minute, and ledger DSR then certifies noise whenever the search happens to stop at the wrong moment. The same agents, when asked for robustness, show better statistical judgement than the metric they are scored by. A sealed holdout makes validity independent of both the agent’s judgement and its stopping time. A holdout that only says PASS or FAIL can be reused by any agent with a bar no harsher than DSR’s. A holdout

that prints Sharpe ratios can be corrected only by a span bound that pays for the full dimension of the strategy space. We recommend that agentic research frameworks expose holdouts only through sealed, budgeted verdicts, and that published agentic backtests report the channel through which the agent saw the holdout, not just the number of trials.

Code and data availability. All code, raw results (including every LLM transcript and RL ledger) and the data pipeline are available at <https://github.com/shloksobti/deflate-by-bits>. Return data are from the Ken French Data Library (CRSP 2026-08 vintage).

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A Proofs

Proof of Theorem 1. Let $\bar{r} = \frac{1}{n} \sum_t r_t$ and $\hat{\Sigma}$ the sample covariance. Then $Z_i = \sqrt{n} \bar{r}_i / \hat{\sigma}_i$, and by the CLT and $\hat{\Sigma} \rightarrow \Sigma$ a.s., $\sqrt{n} \bar{r} \Rightarrow \mathcal{N}(0, \Sigma)$, so $Z \Rightarrow Z^* \sim \mathcal{N}(0, \Sigma)$. Represent $Z^* = \sqrt{\rho} F \mathbf{1} + \sqrt{1-\rho} \varepsilon$ with $F \sim \mathcal{N}(0, 1)$ and $\varepsilon \sim \mathcal{N}(0, I_k)$ independent. Because $\text{med}(Z^*) = \sqrt{\rho} F + \sqrt{1-\rho} \text{med}(\varepsilon)$, the sign vector $s(Z^*) = \text{sign}(\varepsilon - \text{med} \varepsilon)$ is free of F . For even k (ties have probability zero) exactly $k/2$ entries are $+1$, so $\mathbf{1}^\top s = 0$.

The reported portfolio $\hat{w} = s/k$ has $\sqrt{n} \hat{w}^\top \bar{r} = \frac{1}{k} s^\top (\sqrt{n} \bar{r})$ and sample variance $\hat{w}^\top \hat{\Sigma} \hat{w}$. The map $x \mapsto s(x)$ is continuous at Z^* almost surely, so by the continuous-mapping theorem and $\hat{\Sigma} \rightarrow \Sigma$,

$$t_H(\hat{w}) = \frac{\frac{1}{k} s^\top \sqrt{n} \bar{r}}{\sqrt{\hat{w}^\top \hat{\Sigma} \hat{w}}} \Rightarrow \frac{\frac{1}{k} s^\top Z^* \cdot \sigma}{\sqrt{s^\top \Sigma s/k}},$$

where we used $\sqrt{n} \bar{r} = \text{diag}(\hat{\sigma}) Z$ with $\hat{\sigma}_i \rightarrow \sigma = 1$. Now $s^\top \Sigma s = (1-\rho) \|s\|^2 + \rho (\mathbf{1}^\top s)^2 = (1-\rho)k$ and $s^\top Z^* = \sqrt{1-\rho} s^\top \varepsilon = \sqrt{1-\rho} \sum_i |\varepsilon_i - \text{med} \varepsilon|$, using $s^\top \mathbf{1} \text{med} \varepsilon = 0$. Hence $t_H(\hat{w}) \Rightarrow k^{-1/2} \sum_i |\varepsilon_i - \text{med} \varepsilon|$. Finally $\text{med} \varepsilon \rightarrow 0$ a.s. and $\frac{1}{k} \sum_i |\varepsilon_i - m|$ is 1-Lipschitz in m , so $\frac{1}{k} \sum_i |\varepsilon_i - \text{med} \varepsilon| \rightarrow \mathbb{E}|\varepsilon_1| = \sqrt{2/\pi}$ a.s., which gives $t_H(\hat{w})/\sqrt{2k/\pi} \rightarrow 1$.

For $\rho = 0$ and $s_i = \text{sign}(Z_i)$ the same argument gives $s^\top \Sigma s = k$ and $s^\top Z^* = \sum |\varepsilon_i|$, with the same limit. For keep-winners, $\hat{w} = \mathbf{1}\{Z > 0\}/|S|$ with $|S|/k \rightarrow 1/2$, mean $\sum_{i \in S} \varepsilon_i/|S|$ and variance $1/|S|$. The limit is $|S|^{-1/2} \sum_{i \in S} \varepsilon_i \approx \sqrt{k/2} \mathbb{E}[\varepsilon | \varepsilon > 0] = \sqrt{k/2} \sqrt{2/\pi} = \sqrt{k/\pi}$. $\square$

Proof of Corollary 2. The ledger's per-period Sharpe estimates are $Z_i/\sqrt{n}$ for $i \leq k$ together with $t_H(\hat{w})/\sqrt{n}$. Their sample variance satisfies $nV = \frac{1}{k} [\sum_i (Z_i - \bar{Z})^2 + (t_H(\hat{w}) - \bar{Z})^2] (1 + o(1))$. The first term tends to $(1-\rho)$ (the cross-sectional variance of Z^* given F) and the second to $(2k/\pi)/k = 2/\pi$. With Gaussian returns $\hat{\gamma}_3 \rightarrow 0$ and $\hat{\gamma}_4 \rightarrow 3$, so the denominator of (1) tends to $\sqrt{1 + \widehat{\text{SR}}^2/2} = 1 + O(k/n)$. The numerator is $\sqrt{n-1}(\widehat{\text{SR}} - \widehat{\text{SR}}_0) = t_H(\hat{w}) - \sqrt{nV} \mathbb{E} \max_{k+1} + o(1)$, and $\mathbb{E} \max_{k+1} \leq \sqrt{2 \ln(k+1)}$. Therefore the DSR z -score is at least $\sqrt{2k/\pi}(1+o_p(1)) - \sqrt{1-\rho} + 2/\pi \sqrt{2 \ln(k+1)} \rightarrow \infty$ in probability. $\square$

Proof of Proposition 6. A channel revealing b bits has $|\mathcal{T}| \leq 2^b$. By Theorem 3 and the Gaussian tail $\bar{\Phi}^{-1}(p) = \sqrt{2 \ln(1/p)}(1 + o(1))$ as $p \rightarrow 0$, the bar is $\sqrt{2(b \ln 2 + \ln(1/\alpha))}(1 + o(1)) = \sqrt{2b \ln 2}(1 + o(1))$. The sign-flip analyst's report depends on H only through $(\text{sign } Z_i)_{i \leq k}$, which is k bits, and by Theorem 1 its t is $\sqrt{2k/\pi}(1 + o_p(1))$. Hence, for $b = o(n)$ and in the sequential limit of Theorem 1, the supremum over b -bit analysts of the null $(1-\alpha)$ -quantile of $t_H(\hat{w})$ lies between $0.80\sqrt{b}$ and $1.18\sqrt{b}$. $\square$

Proof of Proposition 5. For any w in the span, Cauchy-Schwarz in the $\hat{\Sigma}$ inner product gives $n(w^\top \bar{r})^2 / (w^\top \hat{\Sigma} w) \leq n \bar{r}^\top \hat{\Sigma}^{-1} \bar{r}$. Under Gaussian H_0 the right-hand side is Hotelling's T^2 , distributed as $\frac{k(n-1)}{n-k} F_{k, n-k}$ [12, 24]. The bound holds simultaneously for every w , hence for any data-dependent report. $\square$

Remark (serial dependence and non-Gaussian rows). With stationary, weakly dependent returns, Theorem 1 holds with Σ replaced by the long-run covariance and a HAC t -statistic. Theorem 3 needs the tail of the null t -statistic to be controlled at level $\alpha/|\mathcal{T}|$. For i.i.d. non-Gaussian rows this follows from self-normalised moderate-deviation results when $c_{\mathcal{T}} = o(n^{1/6})$ and the third moment is finite. For HAC statistics tail accuracy is poorer, and a block-bootstrap calibration of $\bar{G}$ is advisable when $|\mathcal{T}|$ is large.

B E6 run history and disclosures

Isolation. In the reported E6 runs, each agent works in a fresh scratch folder outside the project directory and uses a Python environment (NumPy, pandas, SciPy) that is also outside it. It reaches the holdout only through HTTP to a server process holding the train, validation and test matrices in memory. Each null replicate is drawn from an `os.urandom` seed that exists only in the server’s memory and is written to disk after the agent process has exited. Data access is therefore enforced by construction for the null episodes: even an agent that read every file on the machine could not reconstruct the validation data. Claude Code ran with only the Bash, Read, Write and Edit tools, no MCP servers, no user or project settings, and no web access; each session was capped at 25 minutes.

First batch (superseded). We first ran a batch of 72 Claude episodes with an earlier server that wrote the seed to a file in the results directory at start-up, and with an agent Python environment inside the project directory. An audit of those transcripts found that several Sonnet agents listed the project root or searched the filesystem for the client module, but none read the results directory, the seed files, the source code or the data. To make validity enforced rather than behavioural, all reported Claude results come from a complete rerun under the isolation above. The first batch is in `results/exp6/batch1` and gave qualitatively identical results: no Sonnet exploitation; Opus under *optimize* ran programmatic searches in 7 of 12 episodes, reaching validation t of 11–21 on noise, and ledger DSR certified 6 of them at an intermediate stopping point. Before that batch, four episodes were lost to a server-shutdown bug that prevented their seeds from being recorded, and one pilot episode used a different prompt; these five are excluded and archived.

Open-weight coder. The Qwen3-Coder episodes ran on the GPU server with the earlier server version, which wrote seed files at start-up. A search of all 36 transcripts finds no reference to the seed files, the results or project directories, the server code or the raw data; the agents’ code only calls the client API. The coder never exploited the holdout, and access could only have inflated its overfitting, so this cannot affect the conclusions drawn from it.

Models and software. `claude-opus-5-5` and `claude-sonnet-5-5` via Claude Code 2.1.287 (headless mode), run on 2026-10-02/03; Qwen3-Coder-30B-A3B-Instruct-FP8, Qwen2.5-7B/72B-Instruct and Qwen2.5-1.5B-Instruct served with vLLM 0.23 and trained with TRL 1.14 on one AMD MI325X (ROCm 7.14). Full prompts, transcripts and ledgers are in the repository.

C Additional tables

Table 7: E1 details ($\rho = 0$, 200 replicates): mean validation t / mean test t / P(ledger DSR > 0.95).

analyst	$k = 10$	$k = 50$	$k = 100$	$k = 1000$
best-of- k	1.6 / -0.10 / 0.00	2.3 / -0.03 / 0.00	2.5 / -0.06 / 0.00	3.2 / -0.01 / 0.00
keep winners	1.8 / +0.05 / 0.00	4.0 / +0.02 / 0.29	5.7 / +0.05 / 1.00	17.8 / -0.03 / 1.00
sign flip	2.6 / -0.02 / 0.00	5.7 / -0.06 / 1.00	8.0 / -0.08 / 1.00	25.2 / -0.00 / 1.00
rank long-short	2.4 / -0.02 / 0.00	5.6 / -0.09 / 1.00	7.9 / -0.06 / 1.00	25.2 / -0.01 / 1.00
greedy forward	2.4 / -0.02 / 0.16	6.2 / +0.05 / 0.76	8.9 / -0.01 / 0.95	—